Analytical Cumulative Examination
September 19, 1992

Detailed, in depth discussions of a variety of topics have recently appeared in the journal ***Analytical Chemistry*** over the past three months. The titles of the appropriate articles are given below:

(a) Holographic Spectroscopy: Diffraction from Laser-induced Gratings.

(b) Quantitative Structure-Retention Relationships.

(c) Anomalously High Rare-Earth Abundances in Hawaiian Lavas.

(d) Bioanalytical Applications of Partitioning in Aqueous Polymer Two-Phase Systems.

(e) Ion Chromatography: The State of the Art.

1. Choose one topic from the list above and provide a brief (3 - 4 paragraph) overview of the topic.

2. With respect to the specific article you have chosen, discuss the key features that the author suggests make the topic one of importance to the scientific community.

3. For each topic, the author discusses several limitations of the technique(s) used or under development. Briefly, review these limitations.

4. Each article ends with a conclusion section where the future of the technique under development, or the study in progress is discussed. Briefly summarize the authors predictions and comment on whether you feel they seem reasonable.

5. Briefly discuss **your** overall impression of the logic and arguments used by the author to suggest that their technique is ready to revolutionize the way a particular measurement is made.